AIR COMPRESSOR OPERATIONS MANUAL

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Purpose
This manual defines compressor operating principles, telemetry interpretation, operating limits, reliability indicators, alarm philosophy and AI Copilot knowledge.

Equipment Overview
The air compressor converts mechanical energy into compressed air energy. Compressed air is used for instrument air, control valves, pneumatic actuators, utility services and process applications.

Telemetry Parameters
• Air Pressure
• Air Flow Rate
• Vibration
• Motor Temperature
• Oil Temperature
• Oil Pressure
• Motor Current
• Air Leakage Rate
• Bearing Temperature

Operating Limits
Air Pressure: 7–10 bar (Critical: <6 bar)
Air Flow Rate: 500–1000 Nm^3/hr (Critical: <400 Nm^3/hr)
Vibration: 0–4 mm/s (Critical: >7 mm/s)
Motor Temperature: 60–85°C (Critical: >95°C)
Oil Temperature: 60–80°C (Critical: >90°C)
Oil Pressure: 3–5 bar (Critical: <2 bar)
Motor Current: 100–180 A (Critical: >210 A)
Air Leakage Rate: 0–2% (Critical: >5%)
Bearing Temperature: 50–75°C (Critical: >85°C)

Failure Signatures
Bearing Failure: Bearing Temperature ↑, Vibration ↑, Motor Current ↑
Oil Degradation: Oil Temperature ↑, Oil Pressure ↓, Bearing Temperature ↑
Air Leakage: Air Pressure ↓, Air Flow Rate ↓, Air Leakage Rate ↑

AI Copilot Rules
Failure Probability >80% and RUL <30 Days → Maintenance Work Order
Failure Probability >90% and RUL <15 Days → Reliability Review
Failure Probability >95% and RUL <10 Days → Emergency Intervention